A contract-led case for 20 MW electrolysis

Tier 1 investment appraisal | Daily historical data, 2020–2026

20 MW
electrolyser; zero battery

£29.2m
initial investment

£10.94m/year
economic dispatch, six-year mean

£2.12m/year
sales priority, six-year mean

Recommendation. Advance one 20 MW electrolyser to contract and site appraisal at the reference £7.80/kg netback and 10 t/day acceptance cap. Keep battery investment at zero. The six-year case is positive under both dispatch policies, but sales priority loses money in three years. Confirm the buyer's terms and the operating mandate before committing £29.2m.

1 Client boundary, data and assumptions

The supplied generation portfolio and simulated wind farm form the client's fixed starting assets [1]. Demand limits electricity sales. There are no imports, extra export route or payment for unused power. Financial results use repaired prices in constant 2020 pounds.

Table 1. Input treatment. Replacements and estimate flags are recorded in the cell-level audit [2]. Date counts overlap.

Check	Treatment
Calendar and totals	Align five series to 2,452 UTC days through 17 September 2026. Reconcile fuel output to total generation.
Daily statistics	Rebuild 7 generation, 350 demand and 346 price query endpoints from full-day records. Daily mean MW is multiplied by 24 h.
Gaps and prices	Flag observed, interpolated and proxy values. Remove zero-volume price records, volume-weight active providers, and use monthly CPI to express prices in 2020 pounds [3].

Figure 1 shows why 2023 is a useful low-surplus stress year: it has the fewest surplus days and least surplus in the six complete years. We also optimise a common equipment size over 2020–25 and replay it in each year. These are historical evaluations using known daily conditions. Partial 2026 results are retained in the calculation archive.

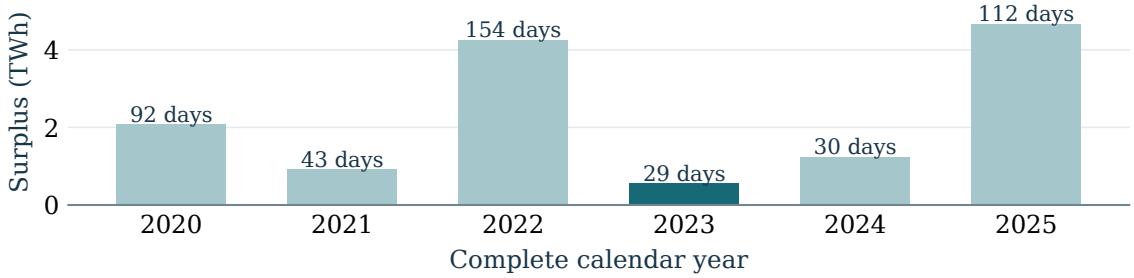


Figure 1. Physical surplus and the number of positive-surplus days. 2023 is the lowest-surplus complete year. Source: repaired daily inputs.

Table 2. Reference technology and contract assumptions, in 2020 pounds.

Input	Assumption and basis
Battery	1.5 MW / 3 MWh modules, 20–80% state of charge, 90% round-trip efficiency, 15-year life. The brief's EUR 288,000/MWh becomes £256,235/MWh. Fixed cost is £14,862/MW-year [1, 4, 8].
Electrolyser	20 MW modules, 20–100% online load, 55.9 kWh/kg. £29.2m installed per module, £0.52m/year fixed cost, 30-year life and £9.79m renewals in years 11 and 22 [5, 6, 8].
Finance and buyer	5% real discount, £7.80/kg netback after non-power variable costs, and a 10 t/day buyer cap. The cap and discount rate are modelling assumptions.

2 Investment model and comparison

2.1 Formulation

For day $t \in \mathcal{T}$, G_t, D_t are supply and demand (MWh), p_t is price (£/MWh), v hydrogen netback (£/kg) and L offtake (kg/day). Decisions q_t, c_t, r_t, h_t, u_t are direct sale, charge, discharge, electrolysis input and unused generation (MWh). Installed b, n and online z_t are module counts. $\kappa = 0.0559$ MWh/kg. Represented years Y equal one for a complete year, six for 2020–25 and 280/365 for the strict 2023 subset.

$$\max \quad \frac{1}{Y} \sum_{t \in \mathcal{T}} [p_t(q_t + r_t) + v h_t/\kappa] - a_B b - a_H n, \tag{1a}$$

$$\text{subject to} \quad q_t + c_t + h_t + u_t = G_t, \qquad q_t + r_t \le D_t, \tag{1b}$$

$$c_t \le E_B b/\eta, \quad r_t \le \eta E_B b, \quad \sum_t r_t = \eta^2 \sum_t c_t, \tag{1c}$$

$$E_H^{\min} z_t \le h_t \le E_H^{\max} z_t, \quad h_t/\kappa \le L, \quad z_t \le n, \tag{1d}$$

$$q_t, c_t, r_t, h_t, u_t \ge 0, \qquad b, n, z_t \in \mathbb{Z}_{\ge 0}. \tag{1e}$$

The day-indexed constraints hold for every t. Equation (1b) allocates generation and limits sales. Equation (1c) gives one usable daily window, $E_B = 1.8$ MWh per battery, with $\eta = \sqrt{0.90}$, and balances storage over the sample. In (1d), $E_H^{\min} = 96$ and $E_H^{\max} = 480$ MWh bound online loading and L limits hydrogen sales. Policy A freely allocates power by value. Policy B also fixes $q_t = \min(G_t, D_t)$; it sells even at negative prices.

Equivalent annual cost for asset $j \in \{B, H\}$ is

$$a_j = F_j + \frac{i}{1 - (1+i)^{-N_j}} \left(K_j + \sum_{k \in \mathcal{R}_j} \frac{C_{jk}}{(1+i)^k} \right). \tag{2}$$

Here F_j, K_j, N_j, i are annual fixed cost, initial cost including commissioning, life and discount rate. $\mathcal{R}_j$ lists renewal years with cost C_{jk}. Gains use the same-policy zero-investment comparator; the code subtracts its constant revenue to resolve small investment differences.

2.2 Reference-year portfolios and matched strategy

Table 3 shows the 2023 stress-year result. The same 20 MW asset under a sale-first rule captures the same 13.92 GWh of surplus but loses £1.07m/year. A earns £9.64m/year more by also diverting saleable electricity. No additional foresight is needed for this no-battery comparison: both use that day's inputs.

Table 3. 2023 capacity (battery / electrolyser MW) and gain (£m/year) under the same-policy comparator.

Scenario	A capacity	A gain	B capacity	B gain
No investment	0 / 0	0.00	0 / 0	0.00
Battery only	0 / 0	0.00	0 / 0	0.00
Hydrogen only	0 / 20	8.57	0 / 0	0.00
Joint choice	0 / 20	8.57	0 / 0	0.00
Fixed 20 MW	0 / 20	8.57	0 / 20	−1.07

Across the complete six-year sample, a common 20 MW size is optimal under both policies. Average incremental values are £10.94m/year in A and £2.12m/year in B. The B outcome depends on the available surplus in each year, as Table 5 shows.

The supported benchmark (£175/MWh, 2012 pounds) becomes £7.80/kg using 39.4 kWh/kg and CPI. We treat this as realised netback with no further non-power deduction, in place of the brief's £9.50/kg. Capital uses the IEA installed midpoint; fixed and stack costs use separate UK reference estimates [7, 5, 6, 8].

3 What changes the investment decision?

3.1 Energy capture and the buyer constraint

Physical surplus is $S_t = \max(G_t - D_t, 0)$. In 2023, unused generation falls from 549.9 GWh to 536.0 GWh. The 20 MW electrolyser uses 171.8 GWh: 13.92 GWh of surplus and 157.9 GWh diverted from direct sale. Surplus is 8.1% of input and the plant captures 2.53% of annual surplus. All 29 surplus days use the full 0.48 GWh/module. At 10 t/day, the buyer can accept at most 0.559 GWh/day of electrolysis input; larger equipment could therefore capture at most 16.21 GWh of that year's surplus.

With no battery and surplus-first attribution, $C_t = \min(h_t, S_t)$. If R_t is wind and solar output, their contribution W_t to capture obeys

$$\max(0, C_t - (G_t - R_t)) \le W_t \le \min(C_t, R_t). \tag{3}$$

The annual interval is 0–13.92 GWh. Adding biomass and run-of-river hydro to the renewable definition leaves this interval unchanged. The aggregate fuel data do not identify the captured source.

3.2 Contract thresholds and retained electricity sales

At 10 t/day, the 20 MW break-even is £4.61/kg in 2023 A and £12.08/kg in B. At £7.80/kg, A first breaks even at 2.23 t/day acceptance; £8.57m/year corresponds to the 10 t/day reference contract. At that cap, doubling equipment to 40 MW reduces annual gain to £7.47m. With 50 t/day, A selects 120 MW and gains £49.09m/year on £175.3m capital.

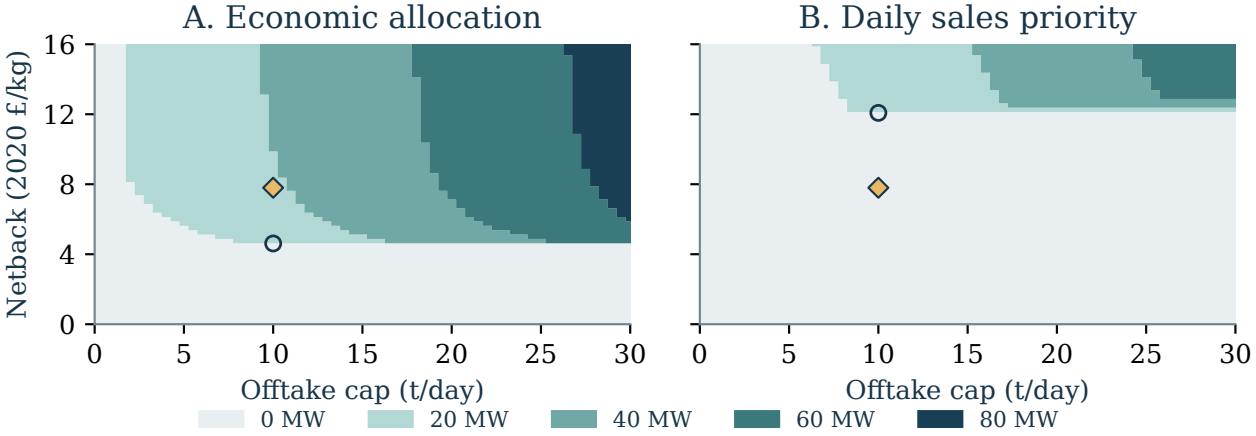


Figure 2. Optimal 2023 capacity. Diamonds are the reference terms; open circles show the fixed 20 MW break-even at 10 t/day. Values after non-power variable costs.

The following annual floors let the operator choose which sales to preserve, relative to 215.1 TWh sold without investment. The 99.99% floor binds. Policy B instead gives electricity priority each day.

Table 4. Realised sales lost and investment under annual floors, 2023 A.

Annual rule	Lost sales (GWh)	H_2 (MW)	Gain (£m/year)
A without floor	157.9	20	8.57
Restore half the sales loss	79.0	20	5.19
Retain 99.99% of sales	21.5	20	1.04
Retain all sales	0.0	0	0.00

Verification. All 60 Xpress schedules reconcile flow and cash; physical residuals are below 0.001 MWh. Eight contract points match enumeration. All 14 annual replays match the independent implementation's published precision [9].

4 Historical performance and investment conditions

Table 5. Fixed 20 MW on all complete years. Gains are £m/year under each policy; dispatch uses realised daily inputs.

Year	Surplus (GWh)	A gain	B gain
2020	2067	16.78	3.12
2021	911	6.29	−0.13
2022	4240	8.91	7.31
2023	550	8.57	−1.07
2024	1229	11.95	−1.00
2025	4660	13.17	4.48
2020–25 mean	2276	10.94	2.12

2023 is a surplus stress case, while 2021 is A's weakest complete year. The positive B mean includes three losing years; a historical average requires the client to tolerate that variation. In the 2025 joint downside of £4/kg and 5 t/day, both policies lose money. A's 2023 gain is still £7.29m/year when initial installation costs rise 25% and the real discount rate becomes 8%, holding stack cost fixed. The strict-quality 2023 subset retains the original capacities.

Battery and model checks. The six-year free-charge revenue ceiling is £54.0k/module-year, below the £96.4k annual cost, ruling out battery investment in that common-size appraisal. For one fixed battery in 2023 A, the annual pool gives £13.3k operating value; chronological stock gives £4.3k, with $s_t = s_{t-1} + \eta c_t - r_t/\eta$, $0 \le s_t \le E_B b$ and zero initial and terminal usable stock.

Table 6. 2023 A cash and annual-value bridge for one module.

Component	£m/year
Hydrogen netback receipts	23.97
Forgone electricity receipts	−12.39
Fixed operating cost	−0.52
Capital and stack equivalent annual cost	−2.49
Equivalent annual gain	8.57

Repeating 2023 operating cash flow for 30 years, including renewals in years 11 and 22 and zero terminal value, gives £131.8m net present value. Payback is 2.64 years simple and 2.90 years discounted. Daily averages omit intraday and ancillary-market value.

5 Recommended next action

1. Contract first. Appraise one 20 MW module against the actual daily acceptance and delivered netback. The reference contract is 10 t/day at £7.80/kg. The 2.23 t/day threshold in 2023 A only marks break-even. **2. Agree dispatch rights.** Use A's sale-diversion option if permitted; retain B and its loss years in the appraisal if daily sales have priority. **3. Stage capital.** Commit £29.2m only after those terms and site access are verified. Larger offtake can justify expansion; keep battery capital at zero under the stated daily service and cost basis.

References

[1] MMCS. *2026/27 Project Brief*, pp. 2–6.
[2] Group 15. *Data repair 20261104-v2*, audit and input series.
[3] ONS. *Consumer Prices Index, D7BT*.
[4] Danish Energy Agency. *Technology data for energy storage*, sheet 180.
[5] IEA. *Global Hydrogen Review 2025*, installed electrolyser costs.
[6] BEIS. *Hydrogen Production Costs 2021* and assumptions annex.
[7] DESNZ. *Hydrogen Allocation Round 1 successful projects*, 2023.
[8] Group 15. *Financial conversions and technology cost basis*, R2–R4 research, 2026, included in the calculation package.
[9] Group 15. *Independent implementation B*, annual replay results, commit `53e3b50`, included in the verification archive.